🧩 Subnetting Part 1 (10.28.2026)

> 📚 **Source:** Jeremy's IT Lab — Day 13 · CCNA 200-301
>
> 🎯 **Focus:** Intro to IPv4 subnetting, especially **Class C** networks
>
> 🧠 **Core idea:** Subnetting = dividing one network into smaller, manageable networks using CIDR/prefix lengths.

🗺️ Big Picture

- **IPv4 address classes** define legacy ranges.
- **CIDR** replaced classful addressing → flexible prefix lengths.
- **Subnetting** borrows host bits to create more networks.
- Each subnet has: **network address**, **usable hosts**, **broadcast address**.
- Devices in different subnets need a **router / Layer 3 device** to communicate.

📦 IPv4 Address Classes

Class	Range	Default Mask	Purpose
🅰️ A	1–126	/8	Large networks
🅱️ B	128–191	/16	Medium networks
🇨 C	192–223	/24	Small networks
🇩 D	224–239	–	**Multicast** ✅
🇪 E	240–255	–	Experimental
🔁 Loopback	127.x.x.x	–	**Loopback**

> ⚠️ **Remember:** Class D = multicast. `127.x.x.x` = loopback.

🧮 CIDR — Classless Inter-Domain Routing

- CIDR = **prefix notation** like `/24`, `/26`, `/30`.
- Replaced old classful rules → allows **any prefix length**.
- Benefits:
  - 🎯 Efficient address allocation
  - 🧩 VLSM — different subnet sizes
  - 📦 Route summarization
- `/26` means: **26 network bits + 6 host bits**.

🔢 Usable Addresses

For a normal subnet:

> **Usable hosts = 2^host bits − 2**

Why minus 2?

- ❌ Network address
- ❌ Broadcast address

Example: `/26` → 6 host bits → `2^6 − 2 = 64 − 2 = 62 usable hosts`.

🧪 Special Prefixes

Prefix	Addresses	Notes
`/31`	2	Point-to-point links; both usable (RFC 3021); no broadcast
`/32`	1	Single host route / loopback-style route

🧠 /26 Example — Class C Split

Class C `/24` = `255.255.255.0`

Split into `/26` = `255.255.255.192`

Block size = **64 addresses** → **4 subnets** per `/24`.

Example: `192.168.1.0/24` → four `/26` subnets:

Subnet	Network	Usable Range	Broadcast
1	`192.168.1.0/26`	`.1 – .62`	`.63`
2	`192.168.1.64/26`	`.65 – .126`	`.127`
3	`192.168.1.128/26`	`.129 – .190`	`.191`
4	`192.168.1.192/26`	`.193 – .254`	`.255`

> 🧠 Memory hook: **/26 = 64-block, 4 subnets per /24, 62 usable each.**

❓ If I Use One /26, What Happens to the Other /26 Subnets?

> 💡 **Short answer:** They are **not automatically used**, **not destroyed**, and **not taken by the first subnet**. They remain **available but dormant** until you configure them.

Example: you assign `192.168.1.0/26` to VLAN 10.

- ✅ `.0/26` is active.
- 😴 `.64/26`, `.128/26`, `.192/26` are **unused address blocks**.
- 🚫 No devices can use them until an interface/VLAN is configured.
- 🚫 No broadcast domain exists for them yet.
- 🚫 Routers will not forward traffic to them unless routes/interfaces exist.
- 📦 They can be assigned later to other VLANs, router interfaces, or networks.
- 🗑️ If never used, that IP space is simply **wasted** — like empty apartments in a building 🏢.

> 🏢 **Analogy:** The `/24` is a building with 4 apartments (`/26`). Renting one apartment does **not** delete the other three. They stay empty and available.

🛠️ Subnetting Process — Basics

1. 📋 Identify requirements: number of subnets + hosts per subnet.
2. 🎚️ Choose prefix length / subnet mask.
3. 🧮 Calculate block size = `2^host bits`.
4. 🗺️ List subnet boundaries.
5. 🧩 Assign subnets to VLANs/interfaces.
6. 🛣️ Configure routing so subnets can communicate.
7. ✅ Verify with ping/traceroute and routing table.

📌 Extra Info to Remember

- **Network address** = first address in subnet.
- **Broadcast address** = last address in normal subnet.
- **Default gateway** must be in the same subnet as the host.
- **Static vs DHCP** = two common address assignment methods.
- Modern networks allow **subnet zero** and **all-ones subnet**; old classful rules sometimes excluded them.
- **VLSM** lets you use different subnet sizes after initial subnetting.
- Subnetting is **administrative** — unused subnets don't "do" anything until configured.

✅ Quick Recap

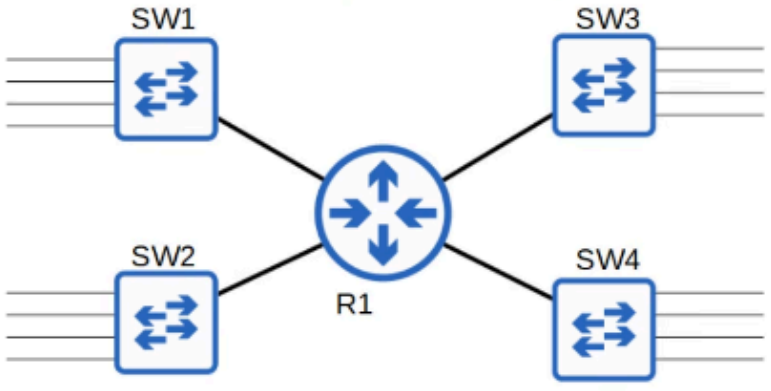
- 🅳 Class D = **multicast**.
- 🔁 `127.x.x.x` = **loopback**.
- 🧮 `/26` = **64 addresses, 62 usable**.
- 🧩 A `/24` becomes **four /26 subnets**.
- 😴 Unused `/26` subnets remain **available but inactive**.
- 🛣️ Different subnets need **Layer 3 routing** to communicate.

🙌 Lesson's Quiz

The first subnet (Subnet 1) is 192.168.1.0/26. What are the remaining subnets?

HINT: Find the broadcast address of Subnet 1. The next address is the network address of Subnet 2. Repeat the process for Subnets 3 and 4.

Subnet 1:
192.168.1.0/26



Subnet 3:
(192.168.1.128)/26

Subnet 2:
(192.168.1.64)/26

Subnet 4:
(192.168.1.192)/26

192.168.1.0/24

My answers are written in red font.